



Plants and nitrogen

Plants use the energy of sunlight to turn carbon dioxide and water into the sugars they need to help them grow. To do this, they also need proteins that contain nitrogen. Although this vital gas makes up two-thirds of the air we breathe, plants cannot absorb nitrogen from the air. Instead, they rely on tiny organisms in the soil to make nitrates both from the nitrogen in the air and from the decaying remains of living things.

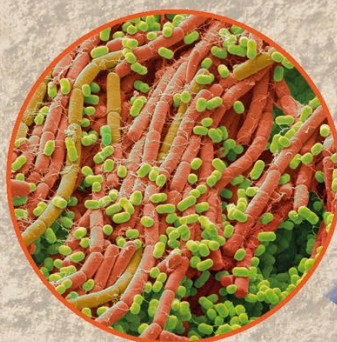
Nitrogen cycle

All plants, animals, and other living things contain nitrogen. When they die, their remains are broken down by fungi and bacteria. This eventually forms nitrates, which plants can use to make proteins that can be eaten by animals. Nitrogen is recycled continuously between the air, soil, and living things in this way—a process called the nitrogen cycle.

1 Nitrogen gas enters the soil from the air. Lightning can also change nitrogen gas into nitrates.

2 Some bacteria in the soil can change nitrogen gas into ammonia, which can be turned into nitrates. Similar bacteria, called nitrogen-fixing bacteria, live in the roots of plants such as peas.

3 Plants absorb nitrates dissolved in the water that their roots soak up from the soil. They use the nitrates to make the proteins essential for growth.



Nitrogen-fixing bacteria

